

VITHURAN SADAGOPAN

Software Development Engineer

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PROFESSIONAL SUMMARY

Engineer on Amazon's global payments platform, the payment-selection and registration widgets in checkout and Prime sign-up across all 21 Amazon retail marketplaces, carrying tens of millions of orders a day with a 99.999 availability. Five years in TypeScript, Node.js, Scala Java and serverless AWS, building the extensibility platform behind 20+ regional payment methods and the developer tooling ~150 engineers ship with. Promoted to SDE II in 2023 and mentored 6 engineers during my journey.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Node.js, Java, Scala, Python, HTML, CSS, GraphQL, SQL, C, C++, Rust

Frameworks & Libraries: React, React Native, VueJS, Svelte, Angular, Django, Vite, Jest, Playwright

Cloud & Infrastructure: AWS Lambda, AWS CDK, AWS Step Functions, DynamoDB, API Gateway, CI/CD pipelines

Practices: Distributed systems, API design, performance optimization, application security, dependency management, CLI tooling

PROFESSIONAL EXPERIENCE

Software Development Engineer II — Amazon

Oct 2023 – Present

Global Payments Experience Platform — payment-selection and registration in checkout and sign-up across all 21 retail marketplaces

- **Onboarded 20+ regional payment methods across all 21 retail marketplaces** by architecting a plugin-based extensibility framework for Amazon's payment UI platform, letting 10 plus partner teams build their registration and purchase experiences in parallel without touching core frontend code.
- **Built the daily development loop for ~150 engineers** spanning my own team, sister teams and teams outside the org — expanded an internal CLI and launched a developer sandbox, turning one-off setup scripts into a full build, test and deploy lifecycle platform
- **Cut server-side-rendering Lambda runtime ~40% and closed an end-of-life security exposure** by driving a Node 14→20 migration across every server-rendering function on the platform, adding memory-utilization monitoring in the process.
- **Narrowed the credential-replay window 24x** — authentication-token TTL cut from 720 to 30 minutes — by driving an authentication-protocol change coordinated across all five sister teams sharing the framework.
- **Removed React rendering bugs and UI jank and reduced rendering latency** by guiding the design and implementation of a JSON-schema form-rendering system built on the Immer state-management library.
- **Caught an upstream checkout-integration failure in production** with a conversion-ratio alarm, and cleared a two-week deployment blocker ahead of Black Friday / Cyber Monday peak.
- **Reduced time to detect upstream client production bugs from 1 week to 1 hour** through AI powered anomaly detection on customer conversion rate alarms

Software Development Engineer I — Amazon

May 2021 – Oct 2023

Payments Experience Platform — web payment portal & registration platform

- **Generated \$300 million in incremental revenue** through modernizing the payments UX in Amazon US marketplace.
- **Cut new payment method onboarding from ~1 month to ~1 week** by building a configurable state machine framework to handle payment method registration workflows. These workflows were further extended to be used in payment method management use cases.
- **Increased reliability of development and production environments by 20%** by building and launching a Web end-to-end testing and canary testing platform to validate customer journeys and authored an onboarding guide for other teams to use.
- **Reduced new-package setup from ~1 week to ~10 minutes** with a single-command developer CLI that auto-configures applications and libraries.
- **Reduced orgs time to prod deploy from 1 week to 1 day** by defining the platform's frontend testing strategy from with coverage reporting piped to business-intelligence dashboards for leadership accountability.

Full Stack Developer (Co-op) — EllisDon, Enterprise Intelligence

May – Aug 2019

- **Shipped a microservice from inception to beta release in 4 months** leading a 4-person team, and standardized project setup with a one-click template (React, Java Spring Boot, Go, MongoDB, Docker).

EDUCATION & CERTIFICATIONS

B.Eng., Computer Engineering (Co-op) — McMaster University, Hamilton, ON

Sept 2017 – Apr 2021

AWS Lambda Advanced (Step Functions & CDK). In progress: AWS Certified Developer – Associate.